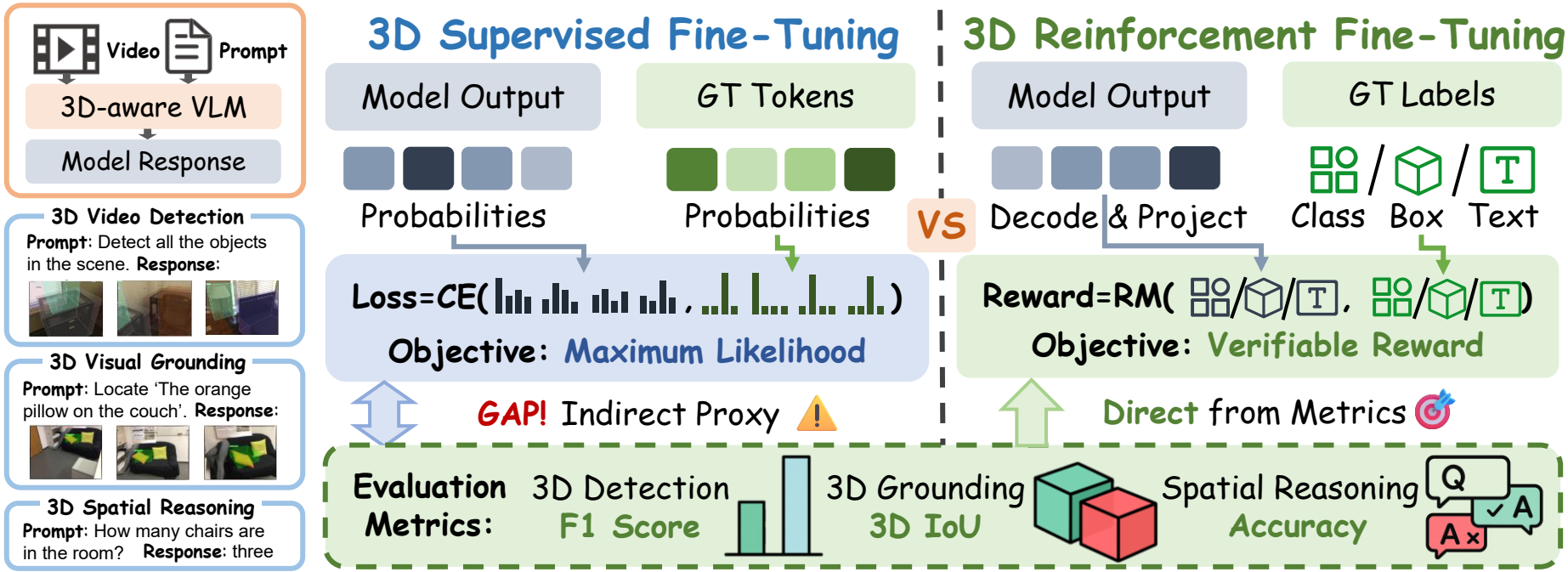
Video
Prompt
3D-aware VLM
Model Response
3D Video Detection
Prompt: Detect all the objects in the scene. Response:
3D Visual Grounding
Prompt: Locate 'The orange pillow on the couch'. Response:
3D Spatial Reasoning
Prompt: How many chairs are in the room? Response: three
3D Supervised Fine-Tuning
Model Output
GT Tokens
Probabilities
Probabilities
Loss=CE(,)
Objective: Maximum Likelihood
GAP! Indirect Proxy
VS
3D Reinforcement Fine-Tuning
Model Output
GT Labels
Decode & Project
Class / Box / Text
Reward=RM(/ / T, / / T)
Objective: Verifiable Reward
Direct from Metrics
Evaluation Metrics:
3D Detection
F1 Score
3D Grounding
3D IoU
Spatial Reasoning
Accuracy
Q
A
A